

NRR-Phi: Text-to-State Mapping for Ambiguity Preservation in LLM Inference

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February 2026

Part of the Non-Resolution Reasoning (NRR) research program.

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Abstract

Ambiguity-bearing inputs often reach downstream systems through interfaces that favor a single resolved response before later context arrives. This creates a representational gap: multiple plausible interpretations may be expressible, but they are not preserved as one manipulable state.

We present a formal framework for *text-to-state mapping* ($\phi : \mathcal{T} \rightarrow \mathcal{S}$) that transforms natural language into a non-collapsing state space where multiple interpretations coexist. The mapping decomposes into three stages: conflict detection, interpretation extraction, and state construction. We instantiate ϕ with a hybrid extraction design that combines rule-based segmentation for explicit conflict markers (adversative conjunctions, hedging expressions) with LLM-assisted enumeration of implicit ambiguity (epistemic, lexical, structural).

On a test set of 68 ambiguous sentences, the resulting states preserve interpretive multiplicity: using hybrid extraction we obtain mean state entropy $H = 1.087$ bits across ambiguity categories, compared to $H = 0$ for single-interpretation collapse baselines used as reference points. We additionally instantiate the rule-based conflict detector for Japanese markers (*kedo*, *kamoshirenai*, etc.) to illustrate cross-lingual portability of the conflict detection stage. This framework extends Non-Resolution Reasoning (NRR) by providing the missing algorithmic bridge between text and the NRR state space, enabling architectural collapse deferment in LLM inference. Design principles for state-to-state transformations are detailed in Appendix D, with empirical validation on 580 test cases (180 single states, 200 contradictory pairs, 200 temporal pairs) demonstrating 0% collapse for principle-satisfying operators versus up to 17.8% for violating operators.

Package surface: The current repository bundles deterministic reruns for the rule-based mapping and operator-validation appendix, together with archived prompt/output artifacts, a transcript-audit path, and a fixed 18-set sanity-check rerun summary for the LLM-handled ambiguity cases reported here.

Keywords: text-to-state mapping, non-resolution reasoning, collapse deferment, ambiguity preservation, LLM inference architecture

1 Introduction

This paper develops the text-to-state mapping layer of the Non-Resolution Reasoning (NRR) framework. It focuses on how competing interpretations are mapped into a retained state, and it

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should be read alongside companion papers that address foundational criteria, implementation, operational calibration, and bounded verification.

1.1 The Early Semantic Commit Problem

Large language models (LLMs) have achieved remarkable capabilities in natural language understanding and generation. However, deployed inference interfaces typically return one resolved response, which creates pressure toward early commitment when context is still incomplete.

Consider the input: “I want to quit my job, but I also don’t want to quit.” A standard one-response interface must produce an output, which pushes the system toward one interpretation of this ambivalent statement. Typical responses either:

- Select one pole of the ambivalence as primary (“It sounds like you’re leaning toward staying...”)
- Reframe the ambivalence as a problem requiring resolution (“Let’s make a pros and cons list...”)
- Request clarification (“Which feeling is stronger?”)

Each of these responses treats the ambivalence as something to resolve. They may enumerate alternatives in text, but they do not by themselves maintain both interpretations as one persistent state for later operators.

This *early semantic commit* pressure is encouraged by common architectural choices in standard LLM design:

Softmax Normalization. Attention mechanisms use softmax to produce probability distributions that sum to unity, forcing interpretations to compete for probability mass rather than coexist independently.

Autoregressive Generation. Each token generation step commits irrevocably to a single choice. Uncertainty must be resolved *before* output, not deferred.

Training Objectives. Cross-entropy loss rewards convergence toward ground-truth tokens, treating maintained ambiguity as prediction error rather than valid epistemic state.

1.2 Non-Resolution Reasoning

Non-Resolution Reasoning (NRR) [1] offers an alternative paradigm that treats ambiguity preservation as a valid computational mode. NRR introduced the state space $\mathcal{S} = \{(v_i, c_i, w_i, m_i)\}$ where multiple interpretations coexist with independent activation weights, and demonstrated that “NRR-lite” implementations ($k=2$ separate embeddings with context gate) maintain high entropy ($H = 0.91$ bits, near-maximum 1.0) at ambiguous turns while baseline models collapse ($H = 0.15$ bits).

However, a critical component remained undefined: **how does text become state?** The mapping $\phi : \mathcal{T} \rightarrow \mathcal{S}$ that transforms natural language input into the NRR state space was described conceptually but not algorithmically specified.

1.3 Contributions

This paper provides the missing algorithmic bridge. Our contributions are:

1. **Formal Definition of ϕ :** We define the text-to-state mapping as a composition of three stages: conflict detection, interpretation extraction, and state construction.

2. **Hybrid Extraction Design:** We combine rule-based segmentation (for explicit conflict markers) with LLM-assisted enumeration (for implicit ambiguity), pairing deterministic executable coverage for explicit-marker cases with archived transcript-audit support for the implicit-ambiguity cases reported here.
3. **Experimental Validation:** We evaluate a hybrid 68-sentence test set, combining three-LLM extraction on 28 English sentences (ChatGPT, Gemini, Claude) with rule-based extraction on the remaining explicit-marker cases, and obtain overall entropy preservation ($H = 1.087$) across ambiguity categories.
4. **Cross-Lingual Portability:** We instantiate the rule-based conflict detector for Japanese markers, demonstrating that the framework is not tied to a single language.

The contribution is algorithmic: this paper shows how text can be mapped into one manipulable retained state, so updates need not default to explicit branch re-enumeration and full comparative checking of every live alternative at each step. It does not prove that retained multiplicity is always cheaper or that future NRR systems have one fixed runtime class.

Support structure. The evidence surface is intentionally layered. Layer 1 is the formal construction showing how text maps into retained state representations. Layer 2 is the current empirical surface: deterministic rule-based reruns, deterministic operator-validation reruns, transcript-audited summaries for the LLM-handled ambiguity cases used in the combined table, and a fixed 18-set current-API sanity-check rerun showing broad family-level stability with one divergent set. A direct semantic-adequacy layer for the LLM-handled cases is not part of the present package and is therefore not claimed as part of the core evidence.

1.4 Paper Organization

Section 2 reviews related work on uncertainty in LLM inference. Section 3 presents the formal framework for text-to-state mapping. Section 4 provides algorithms and implementation details. Section 5 reports experimental results. Section 6 briefly illustrates application to psychological support contexts. Section 7 discusses limitations and future work. Section 8 concludes. Appendix D details the design principles and definitions of the NRR transition operators and metadata functions that transform states while preserving non-collapse.

Scope and series alignment. In this paper, σ_{state} denotes a state-level calibration operator in Appendix D, distinct from the item-level strengthen/dampen operators ($\sigma_{\text{item}}, \delta_{\text{item}}$) used in NRR-IME/Transfer/Coupled. Phi evaluates text-to-state validity and non-collapse properties, especially state entropy and violation rates; later implementation papers add interface and operational metrics such as extraction reliability and token efficiency. These metrics are complementary layers rather than competing definitions of success. Claims are limited to the tested datasets and operator-validation settings, and validity is condition-indexed: a mapped interpretation is valid under its associated context tag, not universally valid across contexts.

2 Related Work

2.1 Uncertainty in LLM Inference

Uncertainty quantification in neural networks has received substantial attention [3, 4]. For LLMs specifically, calibration studies examine whether model confidence aligns with accuracy [5]. However, these approaches focus on *measuring* uncertainty rather than *preserving* it architecturally.

Epistemic uncertainty (model uncertainty) and aleatoric uncertainty (data uncertainty) are typically distinguished [6]. Our framework addresses a different dimension: *interpretive multiplicity*, where multiple valid readings coexist independent of model confidence.

2.2 Multi-Hypothesis Approaches

Beam search and diverse decoding [7] maintain multiple candidates during generation but ultimately select a single output. Mixture-of-Experts architectures [8, 9] route inputs to specialized subnetworks but still produce collapsed outputs.

These approaches treat multiplicity as a means to better single outputs, not as a representational end state. NRR inverts this relationship: non-collapse is the goal, with singular output produced through non-destructive projection when required.

2.3 Ambiguity in NLP

Word sense disambiguation [10] and coreference resolution [11] address specific ambiguity types but assume disambiguation is always desirable. Pragmatic inference models [12] reason about speaker intent but still converge to single interpretations.

2.4 Relation to NRR Framework

This paper builds directly on NRR-Core [1]. That work established the state space formalism and demonstrated entropy preservation through functional verification (NRR-lite achieving $H = 0.91$ bits vs. baseline $H = 0.15$ bits). NRR defines a transition-oriented core operator set $(\delta, \tau, \kappa, \pi)$, with item-level strengthening used in implementations (e.g., NRR-IME), and additional meta-data functions (α, ρ, ι) for representation management (see Appendix D for summary).

This paper fills the gap between text and state space: without a defined ϕ , the NRR operators have no input to operate on. We provide that input transformation.

3 Formal Framework

3.1 Preliminary Definitions

Definition 1 (Text Space). *Let $\mathcal{T}$ denote the space of natural language texts, where each $T \in \mathcal{T}$ is a sequence of tokens $(t_1, t_2, \dots, t_n)$.*

Definition 2 (Semantic State Space). *Following [1], a semantic state space is a structure $\mathcal{S} = (V, C, W, M, \preceq)$ where:*

- $V \subseteq \mathbb{R}^d$ is a set of semantic vector representations
- C is a set of context identifiers
- $W : V \times C \rightarrow [0, 1]$ is an activation weight function
- M is a set of metadata flags (including conflict markers)
- $\preceq$ is a partial order on V (subsumption relation)

A state instance is a set $S = \{(v_i, c_i, w_i, m_i)\}_{i=1}^n$ drawn from this space. Intuitively, a state instance corresponds to a superposition of multiple interpretations, each tagged by its context, weight, and metadata—analogueous to maintaining multiple hypotheses in parallel rather than committing to one.

Definition 3 (Interpretation). *An interpretation of text T is a tuple $\iota = (v, c, w, m)$ where:*

- $v \in V$ is the semantic vector encoding one possible meaning
- $c \in C$ identifies the context under which this interpretation holds
- $w \in [0, 1]$ is the initial activation weight
- $m \in M$ contains metadata (conflict flags, source markers, etc.)

3.2 The Text-to-State Mapping

Definition 4 (Text-to-State Mapping). *The mapping $\phi : \mathcal{T} \rightarrow \mathcal{S}$ transforms input text T to a state space instance:*

$$\phi(T) = \{(v_i, c_i, w_i, m_i) \mid i \in \mathcal{I}(T)\} \quad (1)$$

where $\mathcal{I}(T)$ is the interpretation index set extracted from T .

The mapping ϕ decomposes into three sequential operations:

$$\phi = \psi_{\text{state}} \circ \psi_{\text{interp}} \circ \psi_{\text{conflict}} \quad (2)$$

where:

1. $\psi_{\text{conflict}} : \mathcal{T} \rightarrow \mathcal{T} \times \{0, 1\}^k$ detects conflict markers
2. $\psi_{\text{interp}} : \mathcal{T} \times \{0, 1\}^k \rightarrow \mathcal{I}$ extracts interpretations
3. $\psi_{\text{state}} : \mathcal{I} \rightarrow \mathcal{S}$ constructs the state space

3.3 Stage 1: Conflict Detection

The conflict detection stage identifies linguistic markers indicating multiple coexisting interpretations.

Definition 5 (Conflict Markers). *A conflict marker set $\mathcal{M}_{\text{conf}}$ is a collection of linguistic patterns indicating interpretive multiplicity:*

$$\mathcal{M}_{\text{conf}} = \mathcal{M}_{\text{explicit}} \cup \mathcal{M}_{\text{implicit}} \cup \mathcal{M}_{\text{structural}} \quad (3)$$

where:

- $\mathcal{M}_{\text{explicit}}$: Explicit contradiction markers (e.g., “but”, “however”, “on the other hand”, “yet”)
- $\mathcal{M}_{\text{implicit}}$: Hedging and uncertainty markers (e.g., “maybe”, “perhaps”, “might”, “I think”)
- $\mathcal{M}_{\text{structural}}$: Structural ambiguity indicators (e.g., “both X and Y ”, “either... or”, coordination ambiguity)

The marker set extends to multilingual contexts; for Japanese, corresponding markers include adversatives (*kedo*, *demo*, *shikashi*), hedging expressions (*kamoshirenai*), and epistemic markers (*to omou*). See Appendix for the complete taxonomy.

Definition 6 (Conflict Detection Function). *The conflict detection function ψ_{conflict} returns:*

$$\psi_{\text{conflict}}(T) = (T, \mathbf{f}) \quad (4)$$

where $\mathbf{f} = (f_1, f_2, \dots, f_k) \in \{0, 1\}^k$ is a binary feature vector indicating presence of each marker type.

3.4 Stage 2: Interpretation Extraction

Interpretation extraction transforms text into a set of distinct meaning representations using a hybrid approach.

Definition 7 (Hybrid Extraction). *The interpretation extraction function combines two mechanisms:*

$$\psi_{interp}(T, \mathbf{f}) = \text{Merge}(\psi_{rule}(T, \mathbf{f}), \psi_{LLM}(T, \mathbf{f})) \quad (5)$$

Rule-Based Extraction. For texts with detected conflict markers ($\|\mathbf{f}\|_1 > 0$), rule-based extraction segments text at conflict boundaries:

$$\psi_{rule}(T, \mathbf{f}) = \{(s_j, c_j) \mid j \in J(T, \mathbf{f})\} \quad (6)$$

where $J(T, \mathbf{f})$ indexes segments delimited by conflict markers, s_j is the j -th segment, and c_j is the context label derived from marker type (e.g., “pre-adversative”, “post-adversative”, “hedge-scope”).

LLM-Based Extraction. For complex or implicit ambiguity, LLM-based extraction prompts a language model to enumerate interpretations:

$$\psi_{LLM}(T, \mathbf{f}) = \text{LLM}_\theta(\text{Prompt}_{interp}(T, \mathbf{f})) \quad (7)$$

where Prompt_{interp} is a structured prompt requesting exhaustive interpretation enumeration.

Merge Operation. The merge function removes duplicates via semantic similarity:

$$\text{Merge}(I_1, I_2) = I_1 \cup \{i \in I_2 \mid \forall j \in I_1 : \text{sim}(i, j) < \tau\} \quad (8)$$

where $\text{sim}(i, j)$ is cosine similarity computed over a sentence-level embedding space (e.g., Sentence-BERT [2]), and $\tau \in (0, 1)$ is a conceptual deduplication threshold describing the merge contract. The current paper does not treat one fixed executed τ value as a separately validated package claim.

3.5 Stage 3: State Construction

Definition 8 (State Construction). *Given interpretations $\mathcal{I} = \{(interp_i, ctx_i, conf_i)\}$, the state construction function produces:*

$$\psi_{state}(\mathcal{I}) = \{(v_i, c_i, w_i, m_i)\}_{i=1}^{|\mathcal{I}|} \quad (9)$$

where:

$$v_i = \text{Embed}(interp_i) \in \mathbb{R}^d \quad (10)$$

$$c_i = \text{ContextEncode}(ctx_i) \in C \quad (11)$$

$$w_i = conf_i \cdot (1 + \beta \cdot \mathbf{1}[\text{conflict}]) \quad (12)$$

$$m_i = \{\text{source} : \text{rule}/\text{LLM}, \text{conflict} : \mathbf{f}\} \quad (13)$$

The factor $\beta > 0$ schematically increases activation for interpretations from conflicting contexts, reflecting the NRR principle that conflict indicates information density. In the present paper, β specifies the construction form rather than a separately audited executable package constant.

3.6 Information-Theoretic Properties

Definition 9 (State Entropy). *For a state $S = \{(v_i, c_i, w_i, m_i)\}_{i=1}^n$, the state entropy is the Shannon entropy of the normalized weight distribution:*

$$H(S) = - \sum_{i=1}^n p_i \log_2 p_i, \quad \text{where } p_i = \frac{w_i}{\sum_j w_j} \quad (14)$$

Normalization is applied solely for entropy computation; the weights w_i themselves remain unnormalized in the state representation.

Proposition 1 (Non-Collapse Construction Property of ϕ). *For any text T with $|\mathcal{I}(T)| > 1$ distinct extractable interpretations, the mapping ϕ produces a state S such that:*

1. **(Structural non-collapse)** $|S| \geq 2$, i.e., at least two interpretations are preserved in the state space.
2. **(Entropy preservation)** The state entropy satisfies $H(S) > H_{\text{baseline}}(T)$ where $H_{\text{baseline}}(T) \approx 0$ is the entropy of a collapse-based model that commits to a single interpretation.

Proof. (1) By construction, ψ_{interp} extracts at least two interpretations when conflict markers are detected or when LLM enumeration returns multiple candidates. The merge operation removes only near-duplicates, preserving distinct interpretations.

(2) For collapse-based models, $p^* = 1$ for the selected interpretation, yielding $H_{\text{baseline}} = 0$. For ϕ , with $|S| \geq 2$ and all $w_i > 0$, the distribution has support on multiple elements, guaranteeing $H(S) > 0$. $\square$

Remark 1. *The proposition records a relative construction property: when multiple interpretations are extracted, ϕ preserves more information than collapse-based alternatives. The absolute entropy depends on extraction quality and weight calibration, addressed empirically in Section 5.*

4 Algorithmic Form and Package Surface

4.1 Text-to-State Mapping Algorithm

Algorithm 1 describes the hybrid mapping contract at the design level. The current package exposes deterministic reruns for the explicit-marker rule-based mapping and for Appendix D operator validation. For the implicit ambiguity categories in Section 5, the package exposes archived prompt/output transcripts, a sentence-level manifest, and a local audit script that reconstruct the reported summaries rather than a fully rerunnable live LLM harness.

4.2 NRR Processing Pipeline

The mapping ϕ feeds into the NRR operator pipeline (Appendix D):

In this pipeline, τ is intentionally a no-op on weights: it marks an explicit defer/hold decision instead of forcing an update.

4.3 Computational Complexity

- **Conflict detection:** $O(|T| \times |\mathcal{M}|)$ where $|\mathcal{M}|$ is the marker vocabulary size
- **Rule-based extraction:** $O(|T|)$ for segmentation
- **LLM-based extraction:** Dominated by LLM inference cost
- **State construction:** $O(|\mathcal{I}| \times d)$ for embedding

Algorithm 1 Conceptual Text-to-State Mapping (ϕ)

Require: Text T , embedding function Embed, optional auxiliary interpretation source

Ensure: State $S = \{(v_i, c_i, w_i, m_i)\}$

```
1: // Stage 1: Conflict Detection
2:  $\mathbf{f} \leftarrow \text{DetectConflictMarkers}(T)$ 
3:  $\text{has\_conflict} \leftarrow (\|\mathbf{f}\|_1 > 0)$ 
4: // Stage 2: Interpretation Extraction
5:  $I_{\text{rule}} \leftarrow \emptyset$ 
6: if  $\text{has\_conflict}$  then
7:    $I_{\text{rule}} \leftarrow \text{SegmentAtMarkers}(T, \mathbf{f})$ 
8: end if
9:  $I_{\text{aux}} \leftarrow \text{EnumerateImplicitAmbiguity}(T, \mathbf{f})$ 
10:  $I \leftarrow \text{MergeContract}(I_{\text{rule}}, I_{\text{aux}})$ 
11: // Stage 3: State Construction
12:  $S \leftarrow \emptyset$ 
13: for  $(\text{interp}, \text{ctx}, \text{conf}) \in I$  do
14:    $v \leftarrow \text{Embed}(\text{interp})$ 
15:    $c \leftarrow \text{ContextEncode}(\text{ctx})$ 
16:    $w \leftarrow \text{AssignWeight}(\text{conf}, \text{has\_conflict})$ 
17:    $m \leftarrow \{\text{conflict} : \mathbf{f}, \text{source} : \text{GetSource}(\text{interp})\}$ 
18:    $S \leftarrow S \cup \{(v, c, w, m)\}$ 
19: end for
20: return  $S$ 
```

Algorithm 2 NRR Processing Pipeline

Require: Text T , previous state S_{prev} (or $\emptyset$)

Ensure: Processed state S' , output o

```
1:  $S_{\text{new}} \leftarrow \phi(T)$ 
2:  $S_0 \leftarrow \rho(S_{\text{new}})$  {Optional metadata tagging (weights unchanged)}
3:  $S_1 \leftarrow \tau(S_0)$  {Hold/defer: keep ambiguity as-is for this turn}
4:  $S_2 \leftarrow \delta(S_1, \lambda)$  {Dampen convergence}
5:  $S_3 \leftarrow \kappa(S_2, S_{\text{prev}})$  {Integrate with history (CPP)}
6:  $S' \leftarrow \pi(S_3, S_{\text{prev}})$  {Apply temporal persistence/decay}
7:  $o \leftarrow \Pi(S')$  {Non-destructive projection}
8: return  $(S', o)$ 
```

- **Merge deduplication:** $O(|\mathcal{I}|^2)$ for pairwise similarity

For typical inputs ($|T| < 100$ tokens, $|\mathcal{I}| < 10$ interpretations), the overhead is negligible compared to LLM inference.

5 Experimental Validation

5.1 Research Questions

- **RQ1:** Does ϕ preserve interpretive multiplicity for ambiguous inputs?
- **RQ2:** Which ambiguity categories are captured by rule-based vs. LLM-based extraction?
- **RQ3:** Does the framework generalize across languages (English, Japanese)?
- **RQ4:** How consistent are results across different LLMs (ChatGPT, Gemini, Claude)?

5.2 Test Set

We constructed a test set of 68 ambiguous sentences. For rule-based extraction (adversative, hedging), we include both English and Japanese sentences to demonstrate cross-lingual portability of ψ_{conflict} . For LLM-based extraction (epistemic, lexical, structural), we use English only to isolate the contribution of our framework from the multilingual capabilities of the underlying LLMs.

Category	Total	EN	JP	Method
Adversative	20	10	10	Rule-based
Hedging	20	10	10	Rule-based
Epistemic	8	8	–	LLM-based
Lexical	10	10	–	LLM-based
Structural	10	10	–	LLM-based
Total	68	48	20	

Table 1: Test set composition. Japanese sentences are included only for rule-based categories to demonstrate cross-lingual portability of conflict detection.

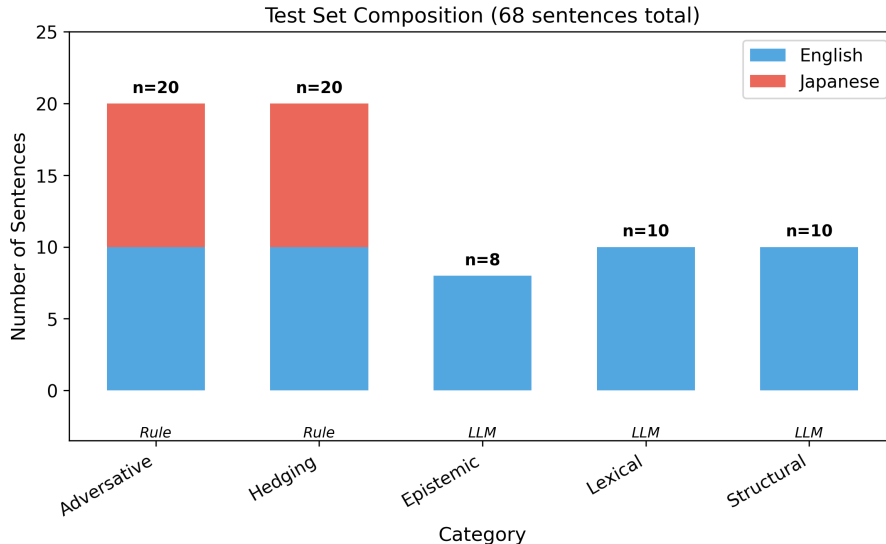


Figure 1: Test set composition across five ambiguity categories (68 sentences total). Rule-based extraction targets adversative and hedging categories with both English and Japanese sentences. LLM-based extraction targets epistemic, lexical, and structural categories with English sentences only.

Adversative sentences contain explicit contradiction markers (“but”, “however”, Japanese *kedo*, *demo*). Example: “I want to quit my job, but I also don’t want to quit.”

Hedging sentences contain uncertainty markers (“maybe”, “perhaps”, Japanese *kamoshirenai*). Example: “Maybe I should apply for that position.”

Epistemic sentences contain belief markers (“I think”, “I believe”). Example: “I think I made the right choice.”

Lexical ambiguity sentences contain polysemous words. Example: “I saw her duck.”

Structural ambiguity sentences have multiple parse trees. Example: “I shot an elephant in my pajamas.”

5.3 Metrics

- **State Size** $|S|$: Number of interpretations in the state space
- **State Entropy** $H(S)$: Shannon entropy of the weight distribution
- **Conflict Detection Rate**: Proportion of sentences where markers were detected

Note that $H_{\max} = \log_2 |S|$ depends on the number of interpretations; $H_{\max} = 1.0$ bit applies only to binary ($|S| = 2$) states. States with $|S| > 2$ can achieve $H > 1.0$ bits.

5.4 Results: Rule-Based Extraction

We first evaluate rule-based extraction alone. Table 2 shows that rule-based methods effectively capture adversative and hedging patterns ($H \approx 1.0$) but fail on epistemic, lexical, and structural ambiguity ($H = 0$).

Category	$ S $	$H(S)$	Method
Adversative	2.10	1.037	Rule-based
Hedging	2.05	1.002	Rule-based
Epistemic	1.00	0.000	Rule-based
Lexical	1.00	0.000	Rule-based
Structural	1.00	0.000	Rule-based

Table 2: Rule-based extraction results. Epistemic, lexical, and structural categories require LLM-based extraction.

Note that Proposition 1 requires $|\mathcal{I}(T)| > 1$ *extractable* interpretations. For epistemic, lexical, and structural categories, rule-based methods cannot extract multiple interpretations ($|\mathcal{I}| = 1$), so the proposition’s precondition is not met. The $H = 0$ results above are not counterexamples but rather motivate the hybrid ϕ that combines rule-based and LLM-based extraction.

5.5 Results: LLM-Based Extraction

To address the limitations of rule-based extraction, we audited archived LLM-generated interpretation sets from three models: ChatGPT, Gemini, and Claude.¹ Each archived run enumerated possible interpretations for 28 English sentences (8 epistemic + 10 lexical + 10 structural). The repository bundles the prompts, saved outputs, and audit manifest used to reconstruct these summaries from `prompts/`.

Key Findings. Within the archived outputs, LLM-assisted extraction recovers interpretive multiplicity for all categories that the current rule-based methods do not cover:

- **Epistemic:** Mean $|S| = 3.25$, $H = 1.673$ (highest entropy, reflecting multiple interpretive layers)
- **Lexical:** Mean $|S| = 2.17$, $H = 1.007$ (captures polysemy)
- **Structural:** Mean $|S| = 2.13$, $H = 0.968$ (captures parse ambiguity)

Within this archived set, Claude extracted the most interpretations on average ($|S| = 3.75$ for epistemic), while all three models yielded $H > 0$ for 100% of test sentences.

¹Experiments used free-tier web interfaces in January 2026; exact model build IDs were not exposed.

Category	ChatGPT	Gemini	Claude	Mean
Epistemic $ S $	3.00	3.00	3.75	3.25
Epistemic $H(S)$	1.577	1.558	1.885	1.673
Lexical $ S $	2.00	2.30	2.20	2.17
Lexical $H(S)$	0.990	1.024	1.008	1.007
Structural $ S $	2.20	2.10	2.10	2.13
Structural $H(S)$	1.056	0.821	1.027	0.968

Table 3: Archived LLM extraction summaries (English only) across three models. All categories yield $H > 0$ in the audited outputs, with epistemic showing the highest entropy due to multiple interpretive layers (assertion vs. hedge vs. self-justification).

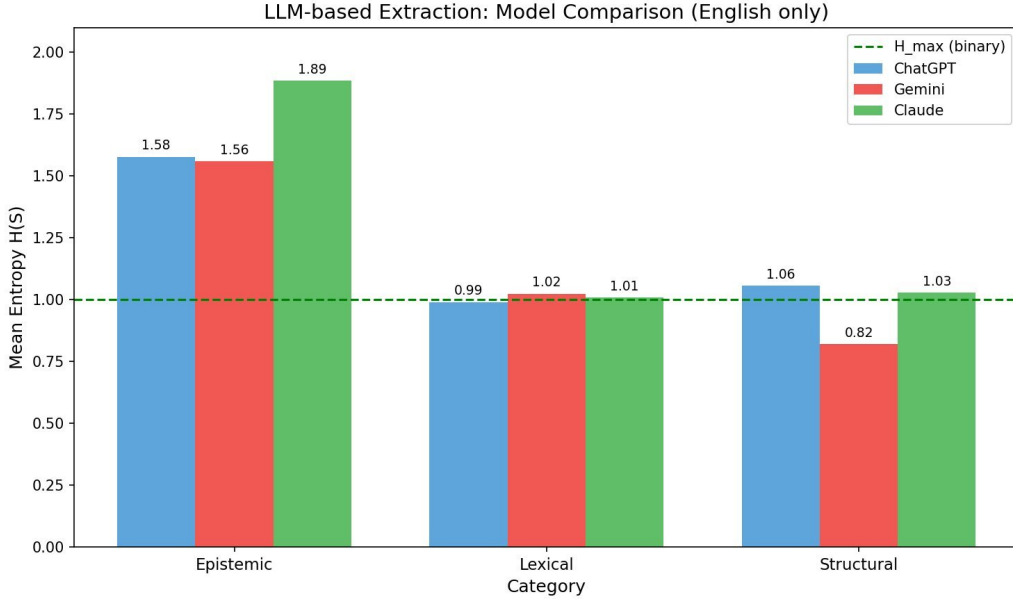


Figure 2: Archived LLM extraction summaries across three models (ChatGPT, Gemini, Claude) for English sentences. The dashed line indicates $H_{\max} = 1.0$ for binary interpretations. All audited model outputs yield $H > 0$ across all categories, with epistemic showing the highest entropy due to multiple interpretive layers.

5.6 Combined Results

Table 4 presents the combined results surface, combining rule-based extraction (for adversative and hedging, including Japanese sentences) with audited archived LLM extraction summaries (for epistemic, lexical, and structural, English only).

5.7 Research Question Answers

(RQ1) Yes. The mapping ϕ preserves interpretive multiplicity: mean $H = 1.087$ bits across all 68 sentences, compared to $H = 0$ for collapse-based baselines.

(RQ2) Rule-based extraction captures explicit markers (adversative, hedging) with $H \approx 1.0$. For the implicit-ambiguity categories in this paper’s archived LLM set, the audited outputs yield $H = 0.97\text{--}1.67$.

(RQ3) The rule-based conflict detector processes Japanese markers (*kedo*, *demo*, *kamoshirenai*) equivalently to English, demonstrating cross-lingual portability of ψ_{conflict} .

(RQ4) In the archived three-model set, all three LLMs (ChatGPT, Gemini, Claude) yield

Category	N	$ S $	$H(S)$	Method
Adversative	20	2.10	1.037	Rule (EN+JP)
Hedging	20	2.05	1.002	Rule (EN+JP)
Epistemic	8	3.25	1.673	LLM (EN)
Lexical	10	2.17	1.007	LLM (EN)
Structural	10	2.13	0.968	LLM (EN)
Overall	68	2.24	1.087	Hybrid

Table 4: Combined results using the reported hybrid evidence surface. All categories achieve $H > 0$, with overall mean entropy $H = 1.087$ bits. The overall row is computed as the weighted mean across all 68 sentences, where LLM-based H values represent the three-model average for each sentence in the archived set.

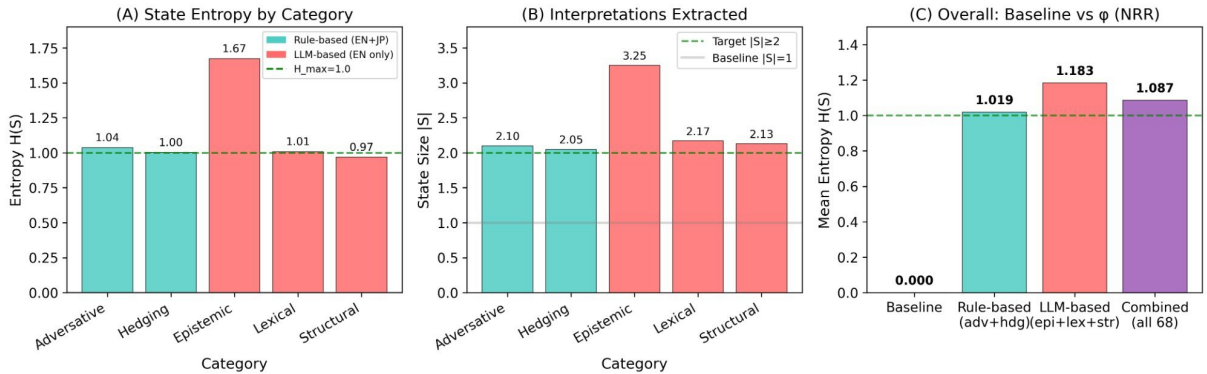


Figure 3: Summary of experimental results. **(A)** State entropy by category and extraction method. Rule-based extraction achieves $H \approx 1.0$ for adversative and hedging; LLM-based extraction achieves $H > 0$ for all categories. **(B)** Mean number of interpretations extracted per category. Epistemic shows highest $|S|$ due to multiple interpretive layers. **(C)** Overall comparison: baseline (single interpretation, $H = 0$) vs. rule-based ($H = 1.019$) vs. LLM-based ($H = 1.183$) vs. combined ($H = 1.087$).

$H > 0$ for 100% of English test sentences. Claude extracts the most interpretations (mean $|S| = 3.75$ for epistemic), while Gemini is more conservative ($|S| = 3.00$). Despite variation in $|S|$, all audited outputs preserve interpretive multiplicity on this dataset.

5.8 Limitations

While the experimental results support the framework, several limitations remain:

- **Test set size:** 68 sentences across 5 categories provides initial validation but larger-scale evaluation is needed.
- **LLM variability:** Different models extract different numbers of interpretations (Claude: 3.75, Gemini: 3.00 for epistemic), suggesting sensitivity to model choice.
- **Human evaluation:** We did not run a full human semantic-adequacy adjudication of the extracted interpretations. The only additional check on the LLM-handled cases is a bounded 18-set current-API rerun, which supports broad family-level stability but not human-labeled correctness.

- **Weight calibration:** Confidence weights from LLMs may not accurately reflect interpretation plausibility.
- **Cross-lingual scope:** We demonstrate Japanese support only for rule-based extraction; full multilingual evaluation requires further work.

Note on LLM role: LLMs serve here as archived sources of interpretation enumeration; the theoretical claims of this paper concern the structure of the resulting states, not the capabilities of the underlying models. The ϕ mapping framework is model-agnostic and could be instantiated with any interpretation source. NRR is a derivation-and-evaluation framework, not a closed recipe; the same derivation lens supports alternative mapping/operator instantiations under fixed protocols with condition-bounded reporting.

6 Illustrative Application: Psychological Support

We briefly illustrate how ϕ applies to psychological support contexts, where ambivalence is common and premature resolution may be counterproductive.

6.1 Motivation

In therapeutic contexts, statements like “I love them, but being with them hurts” express genuine ambivalence that clients benefit from exploring rather than resolving. Off-the-shelf assistant responses often tend to:

- Diagnose the ambivalence as a “problem”
- Offer resolution strategies (pros/cons lists, decision frameworks)
- Select one pole as primary

With ϕ , the ambivalent statement maps to a state with two interpretations:

$$S = \{(\text{“I love them”}, \text{pre-adv}, 0.96, \{\text{conflict} : \text{True}\}), \\ (\text{“being with them hurts”}, \text{post-adv}, 0.96, \{\text{conflict} : \text{True}\})\}$$

Both interpretations are preserved with equal weight, enabling responses that acknowledge the coexistence rather than forcing resolution.

6.2 Limitations

This application is **illustrative, not validated**. Whether NRR-based systems produce therapeutically beneficial outcomes requires:

- Clinical trials with appropriate ethical oversight
- Comparison with established therapeutic approaches
- Long-term outcome measurement

NRR is a computational architecture, not a therapeutic intervention.

7 Discussion

7.1 Implications for LLM Architecture

Natural language is inherently underspecified: identical lexical forms do not guarantee identical referential states due to perspectival variation, temporal-spatial indexicality, and context-dependent conceptualization. Consequently, committing to a single semantic interpretation at inference time is not an information-gain event but an information-loss event. In this view, the role of ϕ is not merely to enumerate alternative readings, but to formalize a separation between the factual layer (what was expressed) and the interpretive layer (what it could plausibly denote), allowing systems to defer semantic collapse until contextual constraints warrant resolution.

The text-to-state mapping demonstrates that early semantic commitment is a *design choice*, not an architectural necessity. Systems can:

1. Detect interpretive multiplicity at input
2. Maintain multiple interpretations through processing
3. Defer collapse until external output is required
4. Preserve internal state after output (non-destructive projection)

In this framing, collapse and commitment are distinct: collapse is a point event at the output boundary, whereas *commit inertia* is trajectory-level persistence of prior selections across turns. The ϕ mapping and non-destructive projection reduce this inertia by keeping alternatives available after each output step. This frame-first design is intentionally structural: by preserving parallel candidate frames after each output step, systems reduce the need for unilateral early forcing in subsequent turns. Accordingly, structural readings are frame-conditional in operation: the same textual surface can map to different relation-structures under different active frame sets, without logical contradiction. We treat this as a condition-bounded architectural implication that downstream papers should test empirically, rather than as a universal interpersonal claim.

This has implications for dialogue systems, where context evolves across turns and early commitments may prove incorrect.

7.2 Distinction from Prompt-Based Enumeration

A natural question arises: if LLMs can enumerate multiple interpretations when prompted, why is ϕ necessary? Note that prompting an LLM to enumerate interpretations results only in a textual list, not in a persistent or manipulable state representation. The model generates these interpretations sequentially and does not maintain them as simultaneously active hypotheses. In contrast, ϕ constructs an explicit state S that can be preserved and transformed by NRR operators across dialogue turns. The distinction is between *generating* ambiguity (textual enumeration) and *maintaining* ambiguity (state persistence). Local judgment can still occur inside the system; the point is that retention itself need not be implemented as repeated full branchwise comparative evaluation during evidence accumulation.

7.3 Limitations

Marker Coverage. The conflict marker taxonomy is not exhaustive. Languages beyond English and Japanese require additional marker sets.

LLM Dependence. LLM-based extraction depends on external model availability and introduces non-determinism. Rule-based extraction is reproducible but limited in coverage.

Weight Calibration. Current weight assignment ($w = \text{conf} \times (1 + \beta)$) is heuristic. Learned weight calibration could improve entropy properties.

Evaluation Scope. The 68-sentence test set covers common ambiguity patterns but may not represent all real-world inputs.

7.4 Future Work

- **Native Non-Collapsing Inference:** Integrate ϕ into LLM architectures rather than as a preprocessing step
- **Learned Conflict Detection:** Train classifiers for marker detection rather than pattern matching
- **Cross-Lingual Extension:** Develop marker taxonomies for additional languages
- **Human Evaluation:** Assess perceived quality of non-collapsing responses in user studies

Reproducibility note. We release deterministic code, fixed settings, and tracked artifacts for the rule-based mapping and operator-validation reruns. For the main-text LLM table, the package releases archived prompts, saved outputs, a sentence-level manifest, and a local audit script that reconstruct the reported summaries rather than replaying live model calls. The full environment specification and artifact map are provided in `nrr-phi/reproducibility.md`.

7.5 Broader Impact and Risk Mitigation

The illustrative psychological-support framing in Section 6 raises asymmetric-risk concerns: a system that preserves ambiguity can still be misread as therapeutic guidance, can surface multiple harmful interpretations without adequate guardrails, or can encourage over-reliance in emotionally sensitive settings. The intended contribution of Phi is therefore architectural and diagnostic, not clinical. Risk mitigation for this line includes keeping the application framing explicitly non-therapeutic, avoiding claims of treatment benefit, pairing any deployment-facing use with human oversight and domain policies, and treating downstream evaluation in sensitive settings as a separate ethics-reviewed task rather than as a consequence of the present results.

8 Conclusion

We have presented a formal framework for text-to-state mapping ($\phi : \mathcal{T} \rightarrow \mathcal{S}$) that transforms natural language into a non-collapsing state space. The mapping decomposes into conflict detection, interpretation extraction, and state construction, with a hybrid approach combining rule-based extraction and LLM-assisted ambiguity enumeration.

The reported evaluation surface shows that the mapping can preserve interpretive multiplicity under the tested extraction conditions: mean state entropy $H = 1.087$ bits across all categories, compared to $H = 0$ for collapse-based baselines. This targets premature collapse, not commitment itself: explicit commitment remains available at the output boundary when conditions support reliable resolution. At the explanatory level, the same explicit state structure supports later separation of retained state from output commitment rather than forcing that distinction to be reconstructed implicitly. In this setup, improvements come from reducing unnecessary early commitment and preserving reversible update paths, not from adding architectural complexity. In the series’ narrow non-evaluative read, Phi formalizes how retained multiplicity can be updated as one structured state rather than by repeated full branchwise comparative evaluation at each step; this is a structural claim about the mapping and update design, not a universal runtime theorem. Unsupported interpretations are not injected at a fixed

evidence state. Hypothesis-set expansion is evidence-gated, unobserved possibilities remain unresolved, and evidence in this paper is restricted to explicit input, provided context, and declared retrieval outputs.

The framework provides the missing algorithmic component of Non-Resolution Reasoning: the transformation from text to the state space on which NRR operators act. In the current package, that contribution is supported by deterministic rule-based and operator-validation reruns together with archived transcript-audit support and a fixed 18-set current-API sanity-check rerun for the implicit-ambiguity cases. The rerun supports bounded use of the archived LLM-handled surface by showing nine consistent sets, eight family-preserving reshapes, and one divergent set, but it does not support perfect identity claims. This is enough to make the bridge explicit and auditable without claiming a fully rerunnable live hybrid implementation for every ambiguity category in the current release.

Early semantic commitment is a design choice. This paper shows how to choose otherwise.

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Category	English	Japanese (romanized)	Context Label
<i>Explicit Contradiction</i>			
Adversative	but, however, yet, although	<i>kedo, demo, shikashi, daga</i>	pre-adv, post-adv
Contrastive	on the other hand, whereas	<i>ippou de, hanmen</i>	contrast-A, contrast-B
Concessive	even though, despite	<i>nimo kakawarazu</i>	concede, main
<i>Implicit Uncertainty</i>			
Hedging	maybe, perhaps, might	<i>kamoshirenai, tabun</i>	hedge-scope
Epistemic	I think, I believe, it seems	<i>to omou, ki ga suru</i>	epistemic-stance
Modal	could, would, should	<i>beki, hazu, darou</i>	modal-world
<i>Structural Ambiguity</i>			
Coordination	both X and Y, either X or Y	<i>mo...mo, ka...ka</i>	coord-A, coord-B
Scope	all...not, every...some	–	wide-scope, narrow-scope

Table 5: Conflict marker taxonomy (English and Japanese)

A Conflict Marker Taxonomy

B LLM Prompt Template

The following prompt template is used for LLM-based interpretation extraction:

Given the text: "[T]"

[If conflict markers detected:

"Note: This text contains potential ambiguity markers."]

List ALL possible interpretations as distinct meanings.

For each interpretation, provide:

1. The interpretation (a clear restatement of one possible meaning)
2. The context/condition under which this interpretation holds
3. Confidence weight from 0.0 to 1.0

Format each as:

INTERP: [interpretation]

CONTEXT: [context]

CONFIDENCE: [0.0-1.0]

C Sample Outputs

This appendix provides representative examples of how ϕ processes different ambiguity types. We show one example each from lexical ambiguity (LLM-based), structural ambiguity (LLM-based), and adversative (rule-based with Japanese).

Example 1: Lexical Ambiguity (LLM-based)

Input: "I saw her duck."

ChatGPT Output:

```
{
  "id": "lex_en_01",
```

```

"interpretations": [
  {
    "meaning": "I observed her pet duck (the bird).",
    "context": "noun-reading",
    "confidence": 0.5
  },
  {
    "meaning": "I saw her lower her head quickly.",
    "context": "verb-reading",
    "confidence": 0.5
  }
]
}

```

State constructed by ϕ : $|S| = 2$, $H(S) = 1.00$ bits

Example 2: Structural Ambiguity (LLM-based)

Input: “I shot an elephant in my pajamas.”

ChatGPT Output:

```

{
  "id": "str_en_01",
  "interpretations": [
    {
      "meaning": "The speaker was wearing pajamas
                  when they shot the elephant.",
      "context": "PP attaches to subject",
      "confidence": 0.7
    },
    {
      "meaning": "The elephant was inside the speaker's
                  pajamas when it was shot.",
      "context": "PP attaches to object",
      "confidence": 0.3
    }
  ]
}

```

State constructed by ϕ : $|S| = 2$, $H(S) = 0.88$ bits

Example 3: Japanese Adversative (Rule-based)

Input: “*Yametai kedo yametakunai*” (“I want to quit, but I also don’t want to quit.”)

Rule-based segmentation: Marker “*kedo*” detected.

Segments extracted:

- Segment 1: “*yametai*” (“I want to quit”) — context: **pre-adv**
- Segment 2: “*yametakunai*” (“I don’t want to quit”) — context: **post-adv**

Weights: Both segments receive $w = 0.5$ (equal weight for adversative).

State constructed by ϕ : $|S| = 2$, $H(S) = 1.00$ bits

These examples illustrate the difference between LLM-based extraction (which enumerates semantic interpretations) and rule-based extraction (which segments based on syntactic markers). Both approaches yield states with $H > 0$, preserving interpretive multiplicity.

D NRR Operator Design Principles and Definitions

This appendix details design principles and formal definitions for NRR’s state-transition machinery. We distinguish (i) core transition operators used for state evolution and (ii) metadata functions used for representation management. While the ϕ mapping (main text) handles text-to-state transformation, these components govern state-to-state processing.

D.1 Design Principles for Non-Collapsing Operators

Transforming NRR states risks destroying the multiplicity that NRR preserves. We identify four principles that operators must satisfy to maintain non-collapse. These principles derive from the structure of the NRR state space itself.

D.1.1 Principle 1: Relative Structure Preservation

Principle 1 (Relative Structure Preservation). *An operator must preserve the proportional relationships between interpretation weights. If $w_i/w_j = r$ before the operation, then $w'_i/w'_j \approx r$ after.*

Rationale: Entropy depends on relative, not absolute, weight values. Transformations that preserve weight *ratios* maintain the entropy structure. Note that ratio preservation is a *sufficient* condition for non-collapse, but not strictly *necessary*—operators such as dampening (δ) preserve non-collapse by increasing entropy, even when ratios are not strictly preserved.

Implementation: Use multiplicative transformations (e.g., $w'_i = \lambda w_i$) or proportional adjustments (e.g., $w'_i = w_i - b \cdot w_i / \max w$) rather than uniform additive changes.

D.1.2 Principle 2: Scale Equivariance

Principle 2 (Scale Equivariance). *An operator’s effect should be independent of the absolute scale of weights. Scaling all weights by a constant should yield the same entropy change.*

Rationale: Since weights are non-normalized, absolute magnitudes are arbitrary. Operations should depend only on relative structure.

Implementation: Design operators such that $\mathcal{O}(\lambda S) \sim \lambda \mathcal{O}(S)$ for scalar $\lambda > 0$. This property is verifiable analytically from the operator definition.

D.1.3 Principle 3: Contradiction Non-Destruction

Principle 3 (Contradiction Non-Destruction). *When merging states with conflicting interpretations, preserve both rather than eliminating one. Contradictions are information, not errors.*

Rationale: This implements the Contradiction-Preservation Principle from NRR foundations. Contradictions are structure, not error; destroying them collapses information.

Implementation: Union with conflict tagging rather than replacement or selection.

D.1.4 Principle 4: Temporal Persistence

Principle 4 (Temporal Persistence). *Interpretations from previous states should persist (with decay) rather than being overwritten, enabling reactivation when context shifts.*

Rationale: Context may shift to favor previously dormant interpretations. Complete overwriting destroys information that may become relevant.

Implementation: Decayed union of current and previous states.

D.1.5 The Information Preservation Law

The principles converge on a single quantitative criterion.

Definition 10 (Information Loss). *The information loss induced by operator $\mathcal{O}$ is:*

$$L(\mathcal{O}, \mathcal{S}) = H(\mathcal{S}) - H(\mathcal{O}(\mathcal{S})) \quad (15)$$

Positive L indicates collapse; negative L indicates information gain.

Definition 11 (Information Preservation Law). *An operator $\mathcal{O}$ satisfies the ϵ -preservation law if:*

$$H(\mathcal{O}(\mathcal{S})) \geq H(\mathcal{S}) - \epsilon \quad (16)$$

for all states $\mathcal{S}$ and small threshold ϵ . Equivalently, $L(\mathcal{O}, \mathcal{S}) \leq \epsilon$.

Operators satisfying Principles 1–4 will satisfy the preservation law; operators violating them risk violation.

D.2 Operator Definitions

We define four core transition operators that have been empirically validated to preserve non-collapse (gD.6), plus one state-level calibration check and three metadata functions.

D.2.1 Core Transition Operators (Empirically Validated)

The following core transition operators have been validated on 180 single states, 200 contradictory pairs, and 200 temporal pairs across three LLMs.

Dampening (δ). Compresses weights toward the mean, reducing dominance of any single interpretation:

$$\delta(\mathcal{S}) = \{(v_i, c_i, w_i(1 - \lambda) + \bar{w}\lambda)\} \quad (17)$$

where $\bar{w} = \frac{1}{n} \sum_j w_j$. Dampening does not strictly preserve weight ratios; rather, it satisfies the *spirit* of Principle 1 by preventing undue suppression of minority interpretations and actually *increases* entropy, thereby satisfying the ϵ -preservation law. Empirically achieves 0% collapse across all tested parameter settings.

State-level Calibration (σ_{state}). Applies a uniform multiplicative scaling to all weights (state-level calibration check):

$$\sigma_{\text{state}}(\mathcal{S}) = \left\{ \left(v_i, c_i, w_i - b \cdot \frac{w_i}{\max_j w_j} \right) \right\} \quad (18)$$

This operation multiplies all weights by the same factor $(1 - b/\max_j w_j)$. Therefore, after normalization for entropy evaluation, the induced probability distribution is unchanged and $\Delta H = 0$. We retain this as a scale-equivariance calibration operator, not as an entropy-changing update rule. This σ_{state} is separate from item-level strengthen operators used in NRR-IME/Transfer/Coupled.

Deferred Resolution (τ). Identity during processing; resolution only at output boundary:

$$\tau(\mathcal{S}) = \mathcal{S} \quad (19)$$

Satisfies all principles by doing nothing until output is required. Serves as baseline in empirical validation.

CPP Integration (κ). Implements Principle 3 (Contradiction Non-Destruction):

$$\kappa(\mathcal{S}, \mathcal{S}') = \mathcal{S} \cup \mathcal{S}' \text{ with conflict tags} \quad (20)$$

Contradictory information is preserved, not eliminated. Validated on 200 contradictory pairs with 0% collapse.

Persistence (π). Implements Principle 4 (Temporal Persistence):

$$\pi(\mathcal{S}_t, \mathcal{S}_{t-1}) = \mathcal{S}_t \cup \{(v_i, c_i, \gamma w_i) : (v_i, c_i, w_i) \in \mathcal{S}_{t-1}\} \quad (21)$$

Previous interpretations persist with decayed activation ($\gamma \in (0, 1)$ is the decay coefficient). Validated on 200 temporal pairs with 0% collapse.

D.2.2 Metadata Functions (Representation Utilities)

The following three functions preserve weights by construction and primarily augment representation metadata rather than perform transition dynamics. They are included as extensible utilities.

Positioning (ρ). Assigns temporal coordinates to distinguish identical tokens across contexts:

$$\rho(\mathcal{S}) = \{(v_i, (c_i, t), w_i)\} \quad (22)$$

Implements contextual identity tracking without affecting weights. Preserves entropy by construction since no weights are modified.

Abstraction (α). Augments state with relational structure:

$$\alpha(\mathcal{S}) = \{(v_i, c_i, w_i, R_i)\} \quad (23)$$

where $R_i = \{d(v_i, v_j) : j \neq i\}$ captures geometric relations. Preserves entropy by construction since no weights are modified.

Invariance (ι). Marks interpretations originating from the same symbol across multiple contexts:

$$\iota(\mathcal{S}_1, \dots, \mathcal{S}_n) = \bigcup_{j=1}^n \mathcal{S}_j \text{ with cross-context ID} \quad (24)$$

where each interpretation $s_i \in \mathcal{S}_j$ receives metadata marking it as derived from the same underlying symbol (e.g., the word “bank” appearing in financial vs. riverbank contexts). By construction, ι retains all interpretations from all input contexts: $|\iota(\mathcal{S}_1, \dots, \mathcal{S}_n)| = \sum_j |\mathcal{S}_j|$, preserving or increasing entropy.

D.2.3 Non-Destructive Projection

For output generation:

$$\Pi(\mathcal{S}) = \arg \max_i w_i, \quad \mathcal{S}' = \mathcal{S} \quad (25)$$

The system produces singular output while preserving internal state. Note that Π is an *output* mechanism, not a processing operator; it is not subject to the design principles, since its purpose is precisely to select a single interpretation for external communication.

D.3 Theoretical Properties

D.3.1 Operator–Data Structure Alignment

Different components require different data structures for validation. Weight-transforming transitions (δ , σ_{state} , τ) operate on individual states and are naturally validated on single states. In contrast, κ (contradiction integration) and π (temporal persistence) are *binary* operators that take two states as input—validating them on single states would be meaningless, as it would fail to exercise their core function. Metadata functions that neither modify weights nor filter interpretations (α , ρ , ι) preserve entropy by construction.

D.3.2 Composition Behavior

Since each principle-satisfying operator individually satisfies $H(\mathcal{O}(\mathcal{S})) \geq H(\mathcal{S}) - \epsilon$, a chain of k such operators accumulates at most $k\epsilon$ information loss. For the transition components in this appendix, this bound is supported analytically: σ_{state} applies an identical multiplicative factor to all weights, so $\Delta H = 0$ exactly (the normalized distribution is invariant); δ moves weights toward the mean, which by the strict concavity of Shannon entropy guarantees $\Delta H \geq 0$; τ is the identity, so $\Delta H = 0$ trivially.

For $\epsilon = 0.1$ bits, a three-component chain $\delta \circ \tau \circ \rho$ is bounded by $\Delta H \leq 0.3$ bits in the worst case. In practice, since δ *increases* entropy and ρ leaves it unchanged, the actual cumulative loss is substantially smaller.

D.3.3 Analytical Verification of Scale Equivariance

Principle 2 (Scale Equivariance) requires that scaling all weights by a constant $c > 0$ does not change the entropy outcome. Since $H(\mathcal{S})$ depends only on normalized probabilities $p_i = w_i / \sum_j w_j$, and scaling all w_i by c leaves p_i invariant, we verify that each operator commutes with scaling:

- σ_{state} (**State-level calibration**): $\sigma_{\text{state}}(c\mathcal{S})_i = cw_i - b \cdot cw_i / c \max w = c(w_i - b \cdot w_i / \max w) = c \cdot \sigma_{\text{state}}(\mathcal{S})_i$. Exact scale equivariance.
- δ (**Dampening**): $\delta(c\mathcal{S})_i = cw_i(1 - \mu) + c\bar{w}\mu = c[w_i(1 - \mu) + \bar{w}\mu] = c \cdot \delta(\mathcal{S})_i$. Exact scale equivariance.
- τ (**Identity**): Trivially equivariant.
- κ (**CPP**): Union of sets; scaling both input states by c scales all weights in the output by c . Equivariant.
- π (**Persistence**): $\pi(c\mathcal{S}_t, c\mathcal{S}_{t-1})$ scales all weights (including decayed ones) by c . Equivariant.
- ρ , α , ι : These operators do not modify weights, so scale equivariance is trivially satisfied.

All core transitions and metadata functions satisfy Principle 2 analytically.

D.4 Operator–Principle Correspondence

Table 6 summarizes which principles each operator satisfies. P1 = Relative Structure Preservation, P2 = Scale Equivariance, P3 = Contradiction Non-Destruction, P4 = Temporal Persistence. Operators marked with $\dagger$ have been empirically validated (§D.6).

Operator	P1	P2	P3	P4
<i>Empirically validated core transitions:</i>				
δ (Dampening) $\dagger$	$\approx^*$	✓	–	–
σ_{state} (Calibration) $\dagger$	✓	✓	–	–
τ (Hold/Defer) $\dagger$	✓	✓	–	–
κ (CPP) $\dagger$	✓	✓	✓	–
π (Persistence) $\dagger$	✓	✓	–	✓
<i>Metadata functions (representation utilities):</i>				
ρ (Positioning) $\S$	✓	✓	–	–
α (Abstraction)	✓	✓	–	–
ι (Invariance)	✓ †	✓	–	–

Table 6: Operator–principle correspondence. $\dagger$ Empirically validated operators (§D.6). $\S\rho$ only adds context metadata and does not modify weights, so entropy preservation holds by construction; empirically confirmed across all validation runs (§D.6). $^*\delta$ does not strictly preserve weight ratios (P1) but satisfies the ϵ -preservation law by *increasing* entropy. $^\dagger\iota$ satisfies P1 by retaining all interpretations from all contexts.

D.5 Pipeline Composition

For implementation, a compact transition chain $\pi \circ \kappa \circ \delta \circ \tau$ is typically sufficient, with ϕ (main text) providing the input transformation from text to initial state. Optional metadata functions (ρ, α, ι) can be inserted where context tracking or cross-context labeling is required. Algorithm 2 shows this compact pattern with optional positioning. Each validated transition preserves the interpretive multiplicity established by ϕ , enabling non-collapsing inference through repeated updates.

D.6 Empirical Validation

We validate that principle-satisfying operators achieve lower collapse rates than principle-violating alternatives. Each operator is validated on the data structure appropriate to its function.

D.6.1 Methodology

Data: Three datasets were constructed using the NRR-Phi ϕ mapping procedure on a **newly constructed sentence set, entirely distinct from the 68-sentence test set in the main text:**

1. **Single states** ($n = 180$): 60 ambiguous sentences (40 epistemic, 20 lexical; English and Japanese) processed by three LLMs (ChatGPT, Gemini, Claude), yielding 180 states. Each state contains 2–4 interpretations with confidence weights.
2. **Contradictory pairs** ($n = 200$): 150 pairs from the same sentence processed by different models (capturing inter-model disagreement; 50 sentences $\times \binom{3}{2} = 3$ model pairs, selected from the 60 sentences by excluding 10 with near-identical model outputs), plus 50 pairs from ambiguous words in different contexts (capturing intra-word polysemy).

3. **Temporal pairs** ($n = 200$): 150 two-turn dialogue pairs (where context evolves across turns), plus 50 context-evolution pairs (where the same sentence is reinterpreted under shifted context).

Metric: Violation rate—percentage of states where $H(\mathcal{S}') < H(\mathcal{S}) - \epsilon$ with $\epsilon = 0.1$ bits. This threshold corresponds to 10% information loss relative to binary states ($H_{\max} = 1.0$ bit).

Operator–data matching: Weight-transforming transitions (δ , σ_{state} , τ) are tested on single states. The integration operator (κ) is tested on contradictory pairs. The persistence operator (π) is tested on temporal pairs. This matching ensures each operator is validated on the data structure it is designed to process.

Comparison: For weight-transforming operators, we compare:

- **Principle-violating:** Uniform subtraction (δ v1: $w'_i = w_i - b$)
- **Principle-satisfying:** Proportional transformation (δ v2: $w'_i = w_i(1 - \lambda) + \bar{w}\lambda$; σ_{state} : $w'_i = w_i - b \cdot w_i / \max w$)

Parameter sweep: δ v1 was tested at three subtraction magnitudes ($b \in \{0.05, 0.10, 0.20\}$), δ v2 at five compression rates ($\lambda \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$), and σ_{state} at four bias levels ($b \in \{0.05, 0.10, 0.15, 0.20\}$). This yields 2,740 total operator–state measurements.

D.6.2 Results

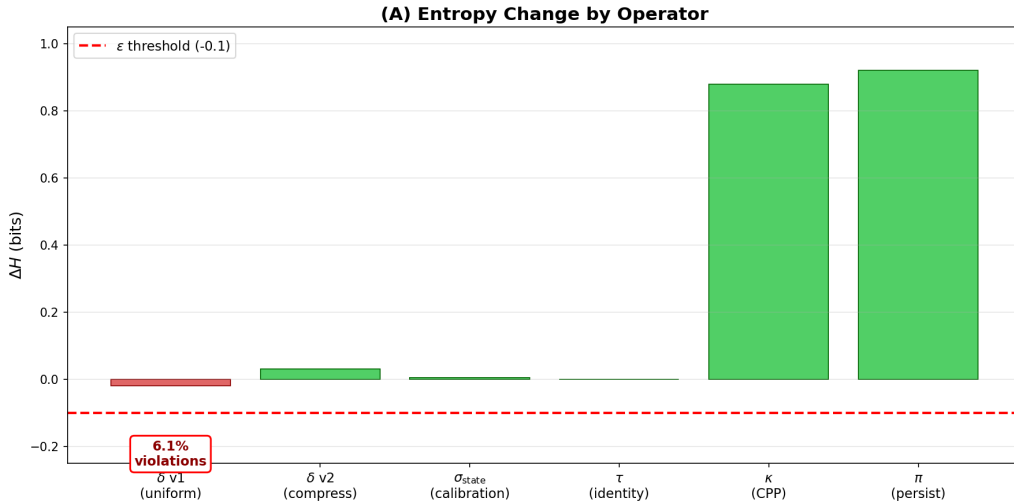


Figure 4: Entropy change by operator type (representative parameter $b = 0.10$ shown for δ v1). The principle-violating uniform operator (δ v1, shown in red) collapses information in 6.1% of cases at $b = 0.10$, falling below the $\Delta H = -\epsilon = -0.1$ threshold (red dashed line); other settings yield 1.7% ($b = 0.05$) and 17.8% ($b = 0.20$). All principle-satisfying operators (δ v2, σ_{state} , τ , κ , π , shown in green) achieve 0% collapse. Operators κ and π substantially increase entropy through integration and persistence mechanisms. The near-zero bars for σ_{state} and τ are expected: σ_{state} is a calibration check, and τ is an intentional hold step.

Figure 4 shows entropy change by operator type. The principle-violating uniform operator (δ v1) collapses information at a rate that increases with subtraction magnitude: 1.7% at $b = 0.05$, 6.1% at $b = 0.10$, and 17.8% at $b = 0.20$ (32 of 180 states). In contrast, all principle-satisfying operators achieve **0% collapse** across all parameter settings.

Table 7 presents the complete results.

Figure 5 presents analysis across the two ambiguity categories from the single-state dataset. Epistemic ambiguities maintain mean entropy $H = 1.81$ bits (SD = 0.49) with an average of

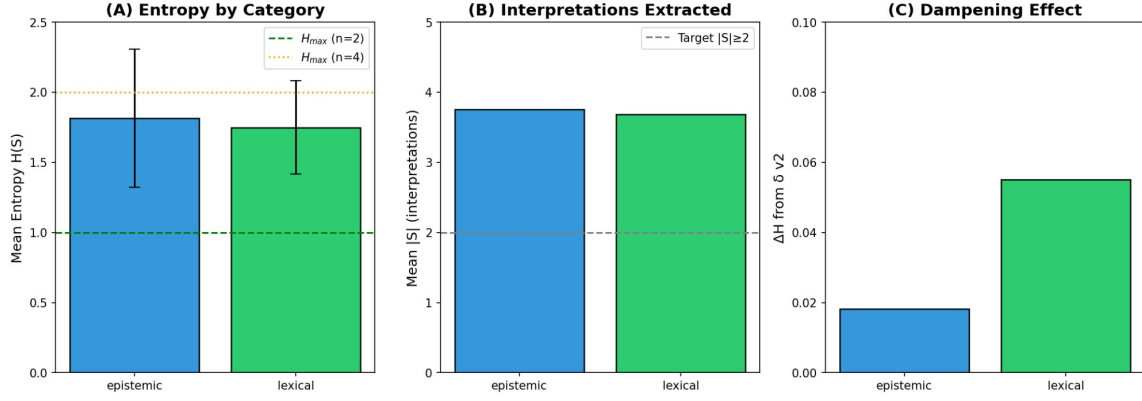


Figure 5: Analysis across ambiguity categories (epistemic, lexical). (A) Mean entropy by category with standard deviation error bars; both categories maintain high multiplicity ($H > 1.7$ bits), approaching the theoretical maximum for four interpretations ($H_{\max} = 2.0$ bits). (B) Average number of interpretations extracted per state; dashed line indicates minimum target ($|S| \geq 2$). (C) Entropy increase from dampening operator ($\delta v2$) relative to pre-dampening state, demonstrating consistent preservation across both categories.

3.76 interpretations per state. Lexical ambiguities show $H = 1.75$ bits (SD = 0.33) with 3.68 interpretations. Both categories demonstrate that the ϕ mapping reliably extracts multiple distinct interpretations.

D.6.3 Key Findings

1. **Collapse rate scales with violation magnitude:** $\delta v1$ violation rate increases monotonically from 1.7% ($b = 0.05$) to 17.8% ($b = 0.20$), confirming that larger uniform subtractions disproportionately suppress weaker interpretations.
2. **Dampening increases entropy:** $\delta v2$ shifts distributions toward uniformity, raising H by an average of 0.03 bits at $\lambda = 0.3$. This increase is analytically guaranteed by the strict concavity of Shannon entropy (see Appendix D) and confirmed empirically. Here, entropy increase denotes redistribution across already extracted interpretations, not insertion of unsupported new interpretations.
3. **State-level calibration is entropy-invariant:** σ_{state} yields mean $\Delta H \approx 0$ bits by design, providing a scale-equivariance check without changing the normalized interpretation distribution.
4. **Integration and persistence add information:** κ and π increase H by +0.88 and +0.92 bits respectively, by merging interpretations from paired states.
5. **Zero violations for all principle-satisfying operators:** Across 180 single states and 400 pairs, no principle-satisfying operator exhibited $\Delta H < -0.1$ bits under any parameter setting tested.

Acknowledgments

The author acknowledges the use of large language models, including Claude (Anthropic), ChatGPT and Codex (OpenAI), and Gemini (Google), for language editing, proofreading, and LaTeX formatting assistance during manuscript preparation. All substantive ideas, claims, analyses, and conclusions are solely the responsibility of the author.

Operator	Data	n	Violations	Mean ΔH
δ v1 ($b = 0.05$)	single	180	3 (1.7%)	-0.012
δ v1 ($b = 0.10$)	single	180	11 (6.1%)	-0.030
δ v1 ($b = 0.20$)	single	180	32 (17.8%)	-0.086
δ v2 ($\lambda = 0.1$)	single	180	0 (0%)	+0.012
δ v2 ($\lambda = 0.3$)	single	180	0 (0%)	+0.031
δ v2 ($\lambda = 0.5$)	single	180	0 (0%)	+0.044
σ_{state} ($b = 0.10$)	single	180	0 (0%)	0.000 [†]
σ_{state} ($b = 0.20$)	single	180	0 (0%)	0.000 [†]
τ (identity, baseline)	single	180	0 (0%)	0.000
κ (CPP)	contra. pairs	200	0 (0%)	+0.881
π (persistence)	temporal pairs	200	0 (0%)	+0.922

Table 7: Experimental validation results (representative parameter values shown; all omitted settings also achieved 0% violations). Principle-violating operators (δ v1) show increasing collapse rates with subtraction magnitude. All principle-satisfying operators achieve 0% violations across all parameter settings tested. κ and π show large positive ΔH because they merge interpretations from two states, increasing multiplicity. [†] σ_{state} applies an identical multiplicative factor $(1 - b/\max w)$ to all weights, leaving the normalized distribution invariant; $\Delta H = 0$ is analytically exact (verified computationally). Similarly, τ (identity) yields $\Delta H = 0$ by mathematical necessity.